

Lecture 3 — Calculus I

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8/20/25

Agenda

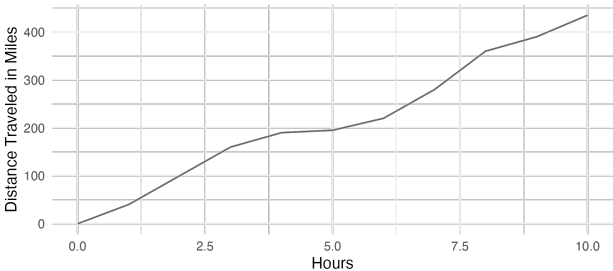
① Differentiation

② Optimization

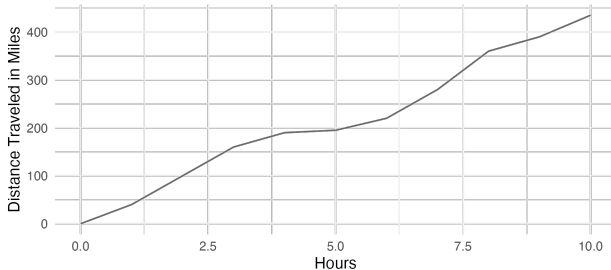
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- ## ① Differentiation

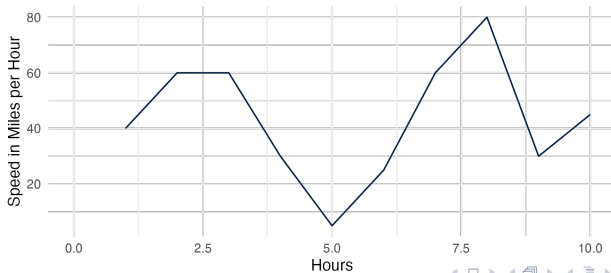
- ## ② Optimization



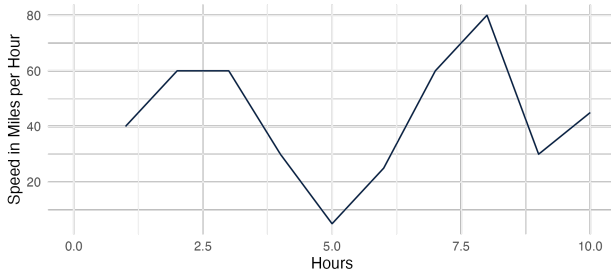
Base Function: Distance Traveled



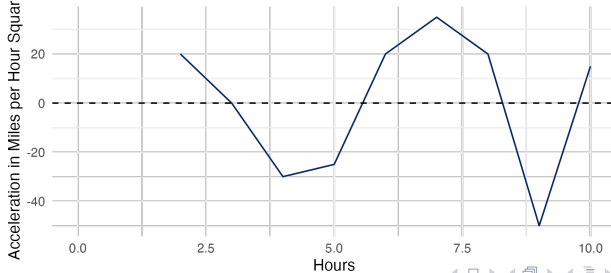
Average Speed per Hour



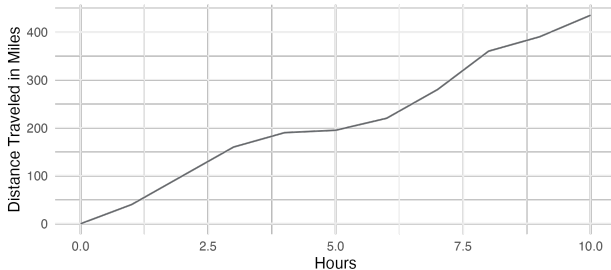
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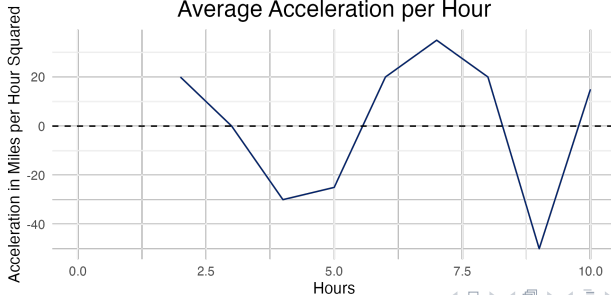
Average Acceleration per Hour



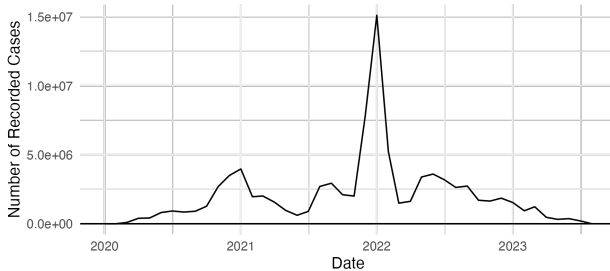
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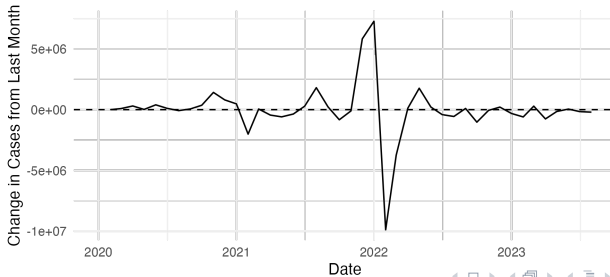
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Monthly Covid Cases



Increase/Decrease in Covid Cases



Types of changes and derivatives

Types of changes:

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Aggregate Heavy Weapons, China

Year	Total	First Difference	Percentage Change
1995	37095	NA	NA
1996	35747	-1348	-3.6%
• 1997	36910	1163	3.3%
1998	37032	122	0.3%
1999	36494	-538	-1.5%
2000	31435	-5059	-13.9%
2001	34281	2846	9.1%

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- Calculated using *limits*

Discrete Change: Secant Lines

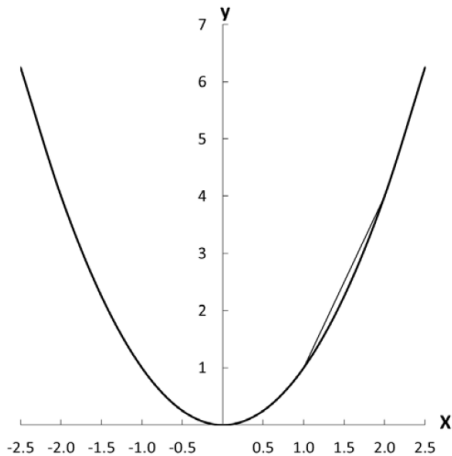


Figure 5.1: Graph of $y = x^2$ with Secant Line

Discrete Change: Secant Lines

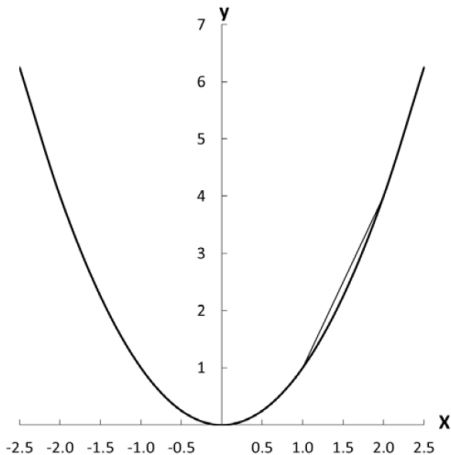


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- Discrete change is computed as
$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

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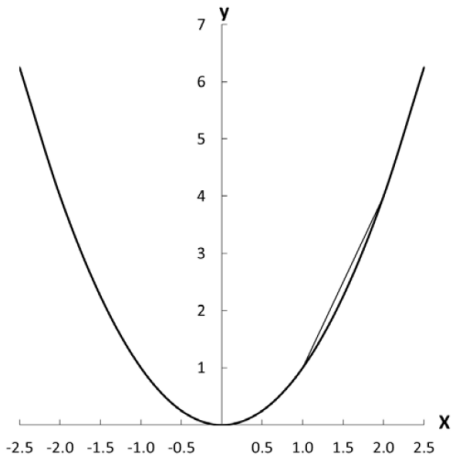


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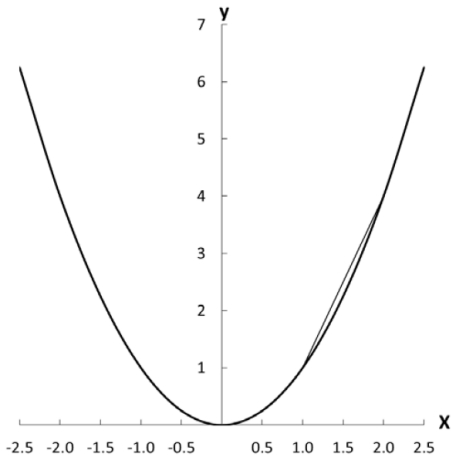


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- Discrete change is the slope of the secant line

Instantaneous Change: Tangent Lines

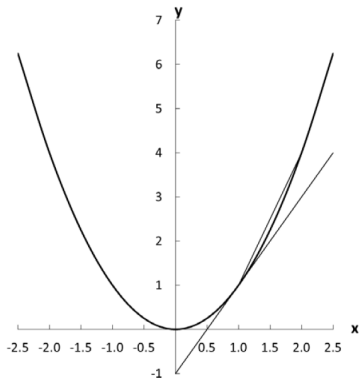


Figure 5.2: Graph of $y = x^2$ with Tangent Line

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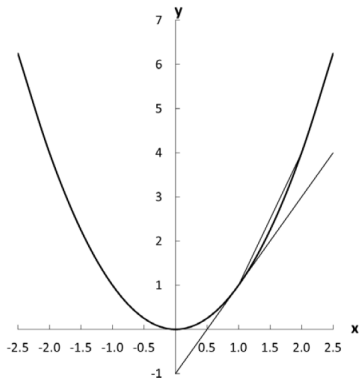


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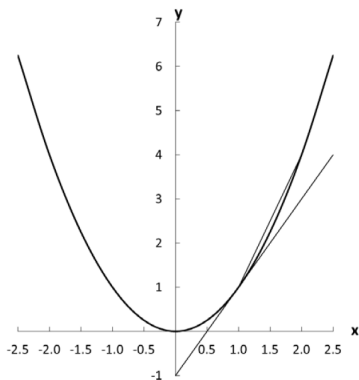


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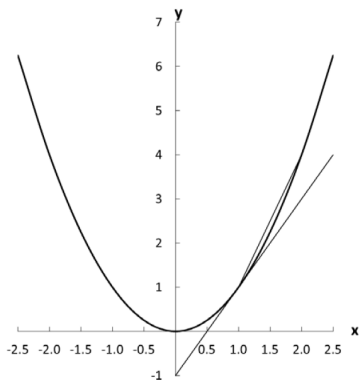


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- Calculate the discrete change for smaller and smaller differences (h) between x_2 and x_1
- $\frac{dy}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Example

Calculate $f'(x)$ using the definition of the derivative:

$$f(x) = x^3 - 16x + 7$$

Solution:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{((x+h)^3 - 16(x+h) + 7) - (x^3 - 16x + 7)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{((x^3 + 3x^2h + 3xh^2 + h^3) - 16x - 16h + 7) - (x^3 - 16x + 7)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - 16h}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 - 16$$

$$f'(x) = 3x^2 - 16$$

Five minutes practice

Calculate $f'(x)$ using the definition of the derivative:

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$$f'(x) = \lim_{h \rightarrow 0} \frac{((x^2 + 2xh + h^2) - 4x - 4h + 3) - (x^2 - 4x + 3)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 2x + h - 4$$

$$f'(x) = 2x - 4$$

Rules of differentiation

Table 6.1: List of Rules of Differentiation

Sum rule	$(f(x) + g(x))' = f'(x) + g'(x)$
Difference rule	$(f(x) - g(x))' = f'(x) - g'(x)$
Multiply by constant rule	$f'(ax) = af'(x)$
Product rule	$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$
Quotient rule	$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
Chain rule	$(g(f(x)))' = g'(f(x))f'(x)$
Inverse function rule	$(f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))}$
Constant rule	$(a)' = 0$
Power rule	$(x^n)' = nx^{n-1}$
Exponential rule 1	$(e^x)' = e^x$
Exponential rule 2	$(a^x)' = a^x(\ln(a))$
Logarithm rule 1	$(\ln(x))' = \frac{1}{x}$
Logarithm rule 2	$(\log_a(x))' = \frac{1}{x(\ln(a))}$
Trigonometric rules	$(\sin(x))' = \cos(x)$ $(\cos(x))' = -\sin(x)$ $(\tan(x))' = 1 + \tan^2(x)$
Piecewise rules	Treat each piece separately

Rules of differentiation Cont

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**Tricky
Differentiation**

Easy Differentiation

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- $f'(x) = 3x^2$

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- $f'(x) = 2x^4 + 4x + 3x^4 + 3x^2$
- $f'(x) = 5x^4 + 3x^2 + 4x$

The Quotient Rule

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- $f'(x) = \frac{(2x^4+4x) - (3x^4+3x^2)}{x^6+4x^3+4}$
- $f'(x) = \frac{-x^4-3x^2+4x}{x^6+4x^3+4}$

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- $f'(x) = (2(3x + 1)) \times 3$
- $f'(x) = 18x + 6$

Examples

- $f(x) = 3x^2$
- $f(x) = \sqrt{x}$
- $f(x) = 2x^9 + x^2 + 8$
- $f(x) = \frac{x^2+5}{x^3+1}$
- $f(x) = (x+5)(x^3 + x^2 + 2)$
- $f(x) = (4x^2 + 2x + 1)^3$
- $f(x) = \ln(2x^4 - x^3 + 3x^2 - 3x)$

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For example:

- Maximization of Utility
- Minimization of Risk
- Maximization of Welfare
- Maximization of the survival probability
- Minimization of errors (for calculating best fit liens)

Extreme value theorem

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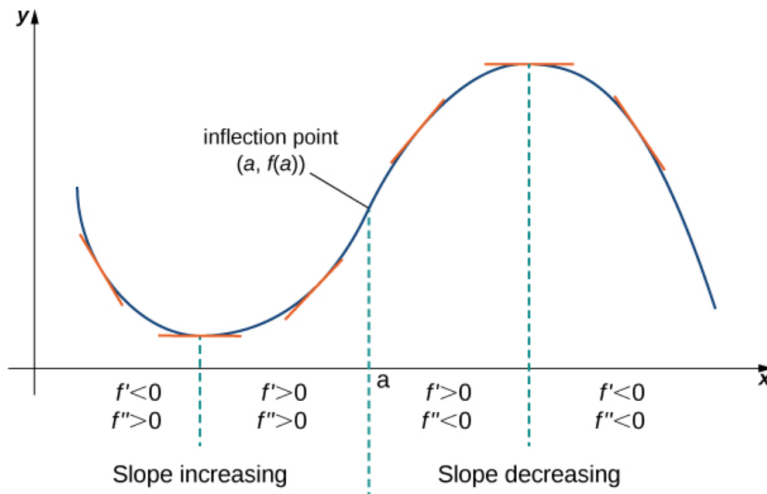
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- A global extremum is the highest (or lowest) point on the function across the full domain

Derivatives and the “shape” of a function



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- ③ Take the derivative of $f'(x)$ to get $f''(x)$
- ④ Second derivative test: Calculate $f''(x^*)$
 - If $f''(x^*) > 0$, x^* is a local minimum.
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- ⑥ If the function is bounded, check the value of $f(x)$ at each bound
- ⑦ Compare the values of $f(x)$ and $f(\text{bounds})$ to find global min/max.

Inflection point or extrema?

- ① If $f'(x^*) = 0$ and $f''(x^*) = 0$, you should continue taking the derivative until $f^n(x^*) =$ a nonzero number.
 - If $n = \text{odd number}$ then x^* is an inflection point, not an extremum.
 - If $n = \text{even number}$ continue to step 2.
- ② Calculate $f^n(x^*)$:
 - If $f^n(x^*) > 0$, the point is a local minimum.
 - If $f^n(x^*) < 0$, the point is a local maximum.

Example optimization!

Find the extrema of the following equation:

$$f(x) = x^3 - 3x^2 + 7, x \in [-4, 4]$$

Also, graph the function.

Example optimization!

1. Take the derivative of $f(x)$ to get $f'(x)$.

$$f(x) = x^3 - 3x^2 + 7$$

$$f'(x) = 3x^2 - 6x$$

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$$f(x) = x^3 - 3x^2 + 7$$

$$f'(x) = 3x^2 - 6x$$

2. First derivative test: Set $f'(x) = 0$ and solve for x^* :

$$f'(x) = 3x^2 - 6x = 0$$

$$3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

Therefore, we will have $x_1^* = 0$ and $x_2^* = 2$

Example optimization!

3. Take the derivative of $f'(x)$ to get $f''(x)$.

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6$$

Example optimization!

3. Take the derivative of $f'(x)$ to get $f''(x)$.

$$\begin{aligned}f'(x) &= 3x^2 - 6x \\f''(x) &= 6x - 6\end{aligned}$$

4. Second derivative test: calculate $f''(x^*)$

- For $x_1^* = 0$: $f''(0) = 6(0) - 6 = -6$.
If $f''(x^*) < 0$, x^* is a local maximum.
- For $x_2^* = 2$: $f''(2) = 6(2) - 6 = 6$.
If $f''(x^*) > 0$, x^* is a local minimum.

Example optimization!

5. Substitute each x^* into $f(x)$ to get (x,y) for each point:

Remember that $f(x) = x^3 - 3x^2 + 7$

- For $x_1^* = 0$: $f(0) = 0^3 - 3(0)^2 + 7 = 7$.
Local maximum at $(0,7)$.
- For $x_2^* = 2$: $f(2) = 2^3 - 3(2)^2 + 7 = 3$.
Local minimum at $(2,3)$.

Example optimization!

5. Substitute each x^* into $f(x)$ to get (x,y) for each point:

Remember that $f(x) = x^3 - 3x^2 + 7$

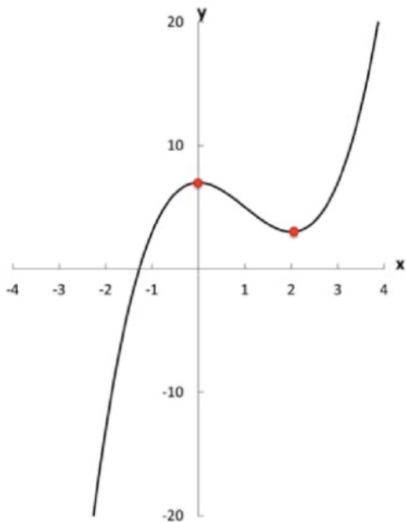
- For $x_1^* = 0$: $f(0) = 0^3 - 3(0)^2 + 7 = 7$.
Local maximum at $(0,7)$.
- For $x_2^* = 2$: $f(2) = 2^3 - 3(2)^2 + 7 = 3$.
Local minimum at $(2,3)$.

6. If the function is bounded, check the value of $f(x)$ at each bound:

Remember that $f(x) = x^3 - 3x^2 + 7, x \in [-4, 4]$

- Lower bound: $f(-4) = (-4)^3 - 3(-4)^2 + 7 = -105$
Global minimum at $(-4, -105)$
- Upper bound: $f(4) = (4)^3 - 3(4)^2 + 7 = 23$
Global maximum at $(4, 23)$.

Example optimization!



7. Compare the values of $f(x^*)$ and $f(\text{bounds})$ to find global min/max:

- Global minimum at $(-4, -105)$
- Local maximum at $(0, 7)$
- Local minimum at $(2, 3)$
- Global maximum at $(4, 23)$